TRANSCRIPT AMD Felix Häse

- Introduction
  - own experiences
  - process of building a pc
- General information about the company
  - secondary sector (manufacturing)
  - designs CPUs, GPUs, accelerators, adaptive SoCs
  - products are used in several sectors e.g. personal computing, HPC, AI infrastructure
  - business model: designing chips + software stacks, system-level architecure, partnering with cloud providers + research institutions e.g. TSMC
- History
  - founded in 1969 by Walter Jeremiah Sanders and seven collegues at Fairchild
  - 1982 → supplying second source chips for Intel (ended 1986)
  - 1991 → released microprocessor compatible with Intels next generation chips (led to legal dispute, which AMD won)
  - early 2000s → release of the first Athlon chip (first chip beeing able to reach a 1 Ghz clock speed)
  - 2011 → major contracts with Sony (PS4) and Microsoft (Xbox One) to produce chips for gaming consoles
  - 2025 → contract with OpenAI to supply chips for expanding data center for ChatGPT
- Organisation model
  - organized around product lines/ operating segments
  - Data Center/ Server (e.g. Threadripper CPUs)
  - Client/ Consumer (e.g. notebooks, Ryzen processors)
  - Gaming/ Graphics (e.g. GPUs, gaming consoles)
  - Research and Development (e.g. microarchitecure, energy efficiency)
- Current major projects
  - instinct MI350 series accelerators: new gen GPUs aimed at generative AI & HPC workloads
  - ROCm 7 software stack: improving developer expierence, expanding hardware compatibility
  - AMD developer cloud: open-source community access to AMD's compute environment
- Future goals
  - Leadership in AI and Data Center (beat current competition Nvidia and Intel)
  - open ecosystem (open rack designs and software stacks)
  - Efficiency leadership (not just raw performance, but performance per Watt)
  → reduce power usage of AI/HPC systems
  - expand markets
  - improve sustainibility

<u>Keywords:</u>
SoC: an integrated circuit that combines the essential components of a computer or electronic system onto a single microchip
HPC: high performance computing → using powerful and interconnected computers to solve complex problems
TSMC: Taiwan Semiconductor Manufacturing Company → worlds largest independent semiconductor manufacturer
software stack: a combination of software components, that work together to build and run an application
Fairchild: was an American semiconductor company, manufacturing and developoing semiconductors